

Homework 6

Due: August 29, 2018

Section	Pages	Numbers
8.2	417-420	1, 3, 7, 13, 19, 21, 23, 25
Additional Problems	N/A	See below

- Solve the following differential equations.
 - $\frac{dy}{dx} = y + x$
 - $x' = 2x + te^{2t}$
 - $(1+t)y' - ty = t + t^2$
 - $tx' + (1+t)x = e^{-t} \sin 2t$
- Let S represent the amount (in pounds) of salt in a tank at time t minutes. A solution containing 2 lbs. of salt per gallon flows into a tank at the rate of 3 gal./min. and the well-stirred mixture flows out of the tank at the rate of 5 gal./min. The tank initially holds 500 gallons of solution containing 25 lbs. of salt.
 - How many gallons of solution are in the tank after 1 minute? After 10 minutes? After 50 minutes? After t minutes? When will the tank be empty?
 - Set up a differential equation with initial conditions representing the rate of change of salt in the tank. Solve the equation.
 - How much salt is in the tank after 10 minutes? After 100 minutes? After 200 minutes? After 240 minutes?
 - What will the maximum amount of salt in the tank and at what time will it occur?